

The Impact of Social Media on Social Connection & Isolation

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AI Tools

- ChatGPT
- Consensus
- Elicit
- Pubmed

Measurement Tools

- health measures
- NIH/PROMIS

Keywords

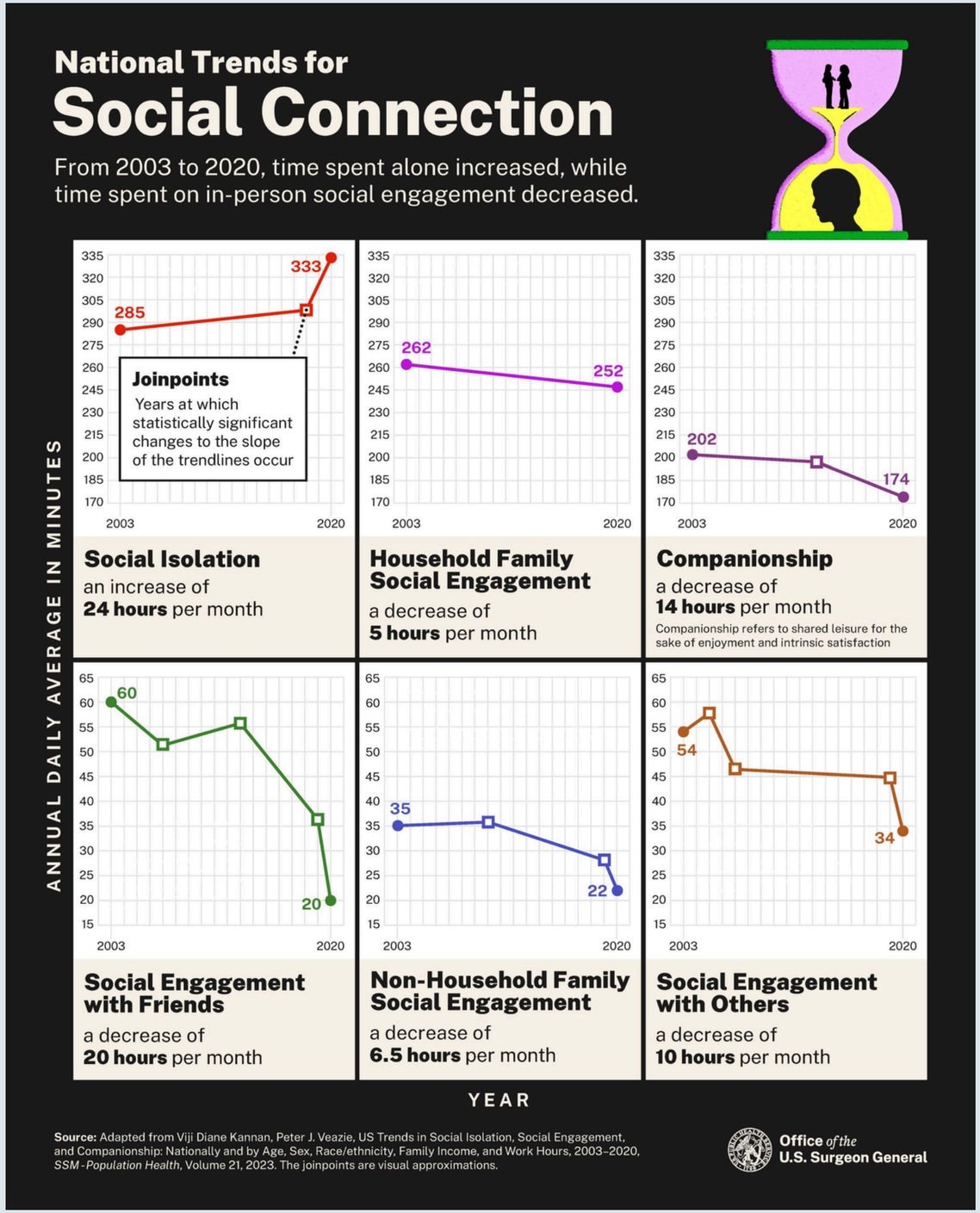
“Social isolation and morbidity/mortality” “Digital and physical social interactions” “Adolescent social media usage and isolation” “Social Network Theory” “Social network density and centrality” “Perceived social isolation in youth” “Social media impact on interpersonal connectivity”

Background

Literature Review

What is the Problem?

- Social media and modern technologies have fundamentally changed how we interact and maintain relationships.
- Despite the intention to connect people, studies show an increase in perceived social isolation.
- Over the past decades, the number of close friendships has decreased. For instance, nearly half of Americans in 2021 reported having three or fewer close friends, a significant drop in proportions from 1990 (Cox, 2021).
- The COVID-19 pandemic exacerbated this trend, with a study noting a 16% decline in social network sizes from 2019 to 2020. (Kovacs et al., 2021)



- **National Trends on Social Isolation adapted from 2003–2020 American Time Use Survey data**
- Time spent alone has risen from 285 minutes per day in 2003 to 333 minutes per day in 2020, marking an increase of 24 hours per month.

In-person social engagement with friends has decreased significantly:

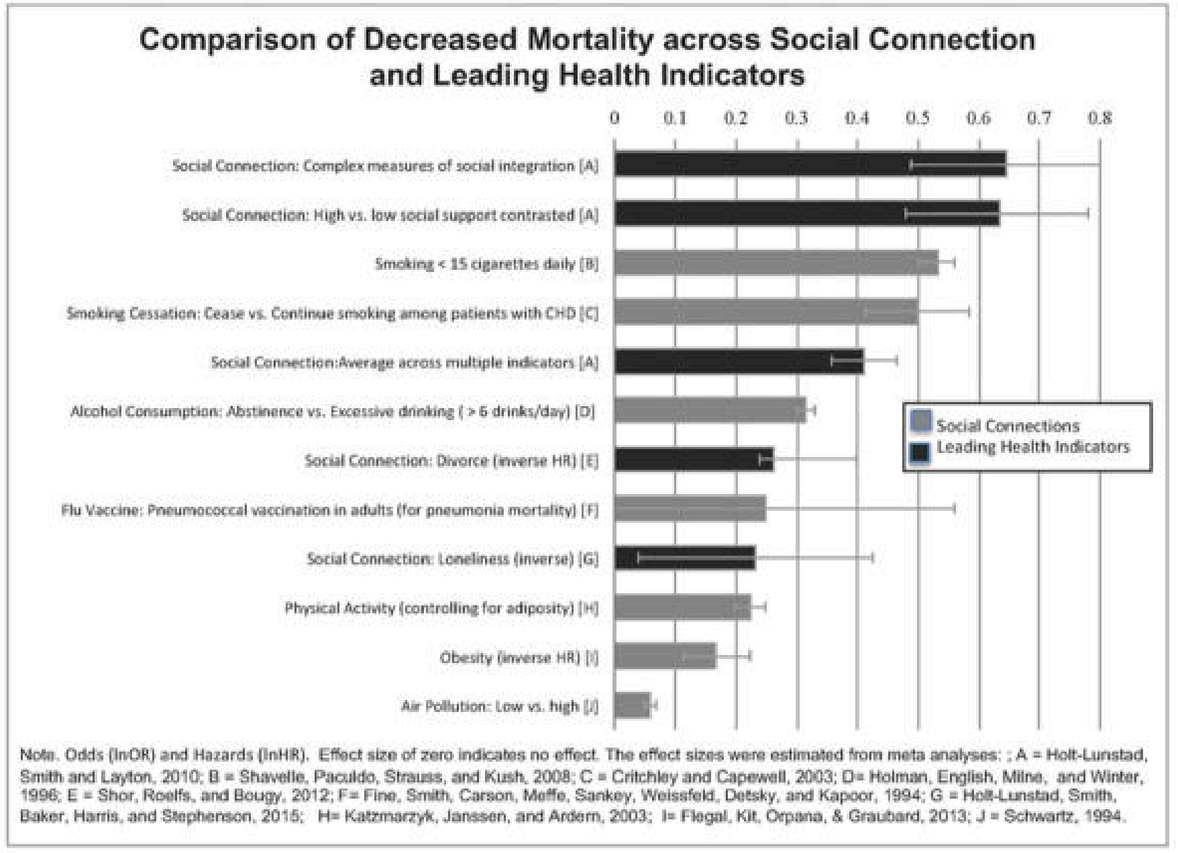
- From 60 minutes per day in 2003 to 20 minutes per day in 2020.
- This is particularly pronounced among **young adults (15-24 years),** where daily in-person time with friends dropped by nearly 70% over two decades (Kannan et al., 2022).

- Growing trend of reduced social intimacy raising concerns about broader impacts on mental health and well-being.

Background
Literature Review

The Effects of Social Isolation

- Social isolation significantly increases the risk of morbidity and mortality, on par with risks associated with smoking and obesity. (Holt-Lunstad et al., 2015)
- Studies reveal that the lack of social connections increases the risk of premature death by up to 50%, which is comparable to smoking 15 cigarettes a day. (Holt-Lunstad et al., 2017)
- Research indicates that social isolation can lead to substantial changes in brain structure and function, affecting areas related to cognition and emotional processing. (Finley & Schaefer, 2022)



Background

Literature Review

Evidence & Literature:

- **Impact on Social Connections:** Research by Twenge et al. (2018) highlights that increased screen time correlates with a decline in face-to-face interactions, contributing to feelings of loneliness.
- **Youth Vulnerability:** Gaspar et al. (2023) found that adolescents engaging in passive social media use (e.g., scrolling without interaction) are more likely to report feelings of loneliness and social disconnection.
- A study by Primack et al. (2017) found that young adults who spent more than two hours per day on social media were twice as likely to report perceived social isolation compared to those who spent less than 30 minutes daily.
- **Older Populations:** Kim et al. (2022) found that specific social network characteristics, such as the density and centrality of ties, predict loneliness in older adults. These findings are relevant for understanding similar dynamics in adolescents who are increasingly reliant on social media for their social connections.

Citations/Sources

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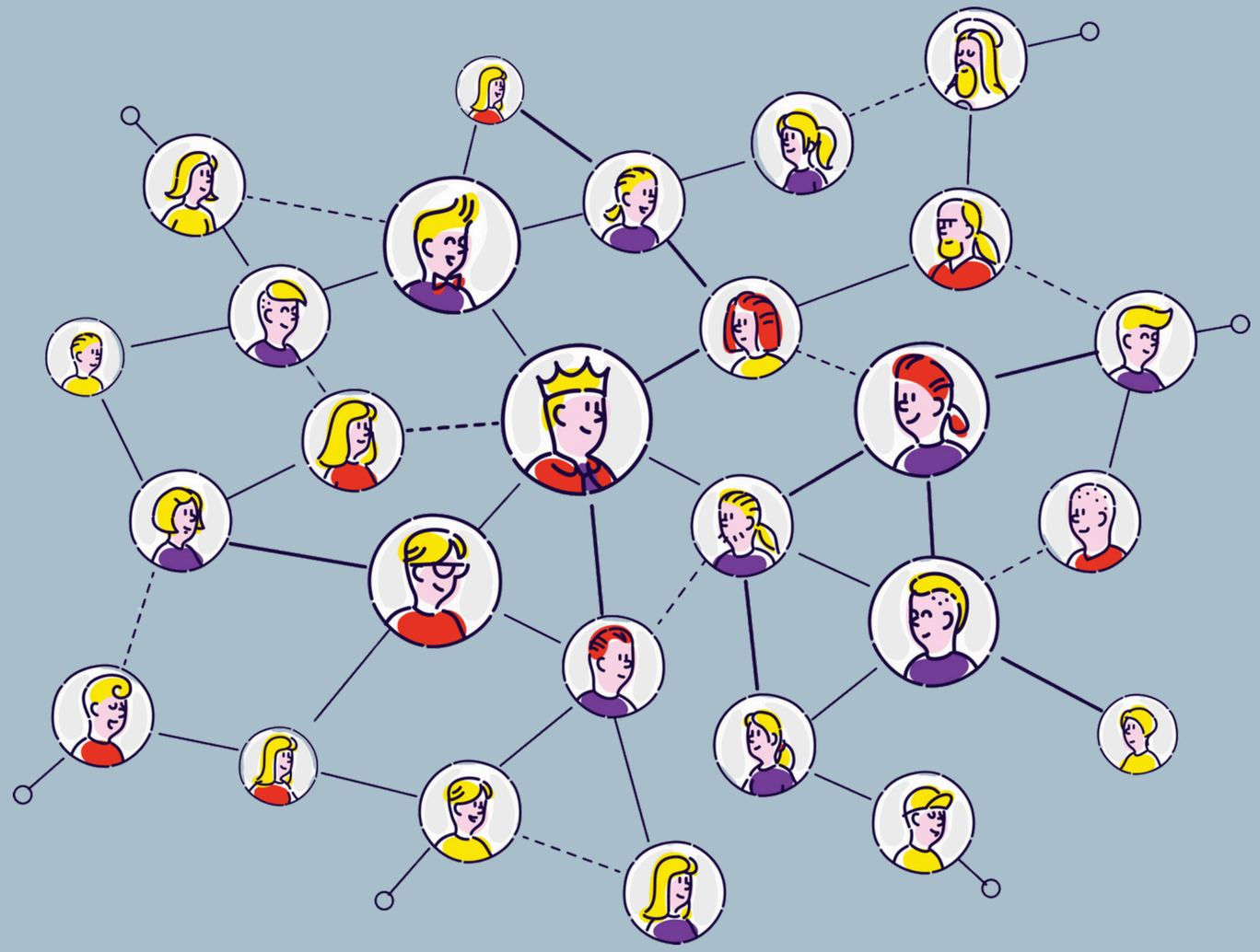
Background (Cont.)

Gaps in Knowledge

What are the gaps in our knowledge and literature?

- **Limited longitudinal research:** There's limited research on the long-term effects of social media on social isolation and network structures, especially concerning adolescents and young adults.
- Lack of existing strategies to mitigate the negative impacts of social media on social isolation while enhancing its positive aspects.
- **Population focus:** Many social network studies focus on elderly populations, with less data available on social isolation SNA in adolescents and young adults.
- Addressing gaps is crucial for developing targeted interventions that promote healthy social connections in the digital age, particularly for vulnerable populations such as adolescents.

Social Network Theory



Background (Cont.)

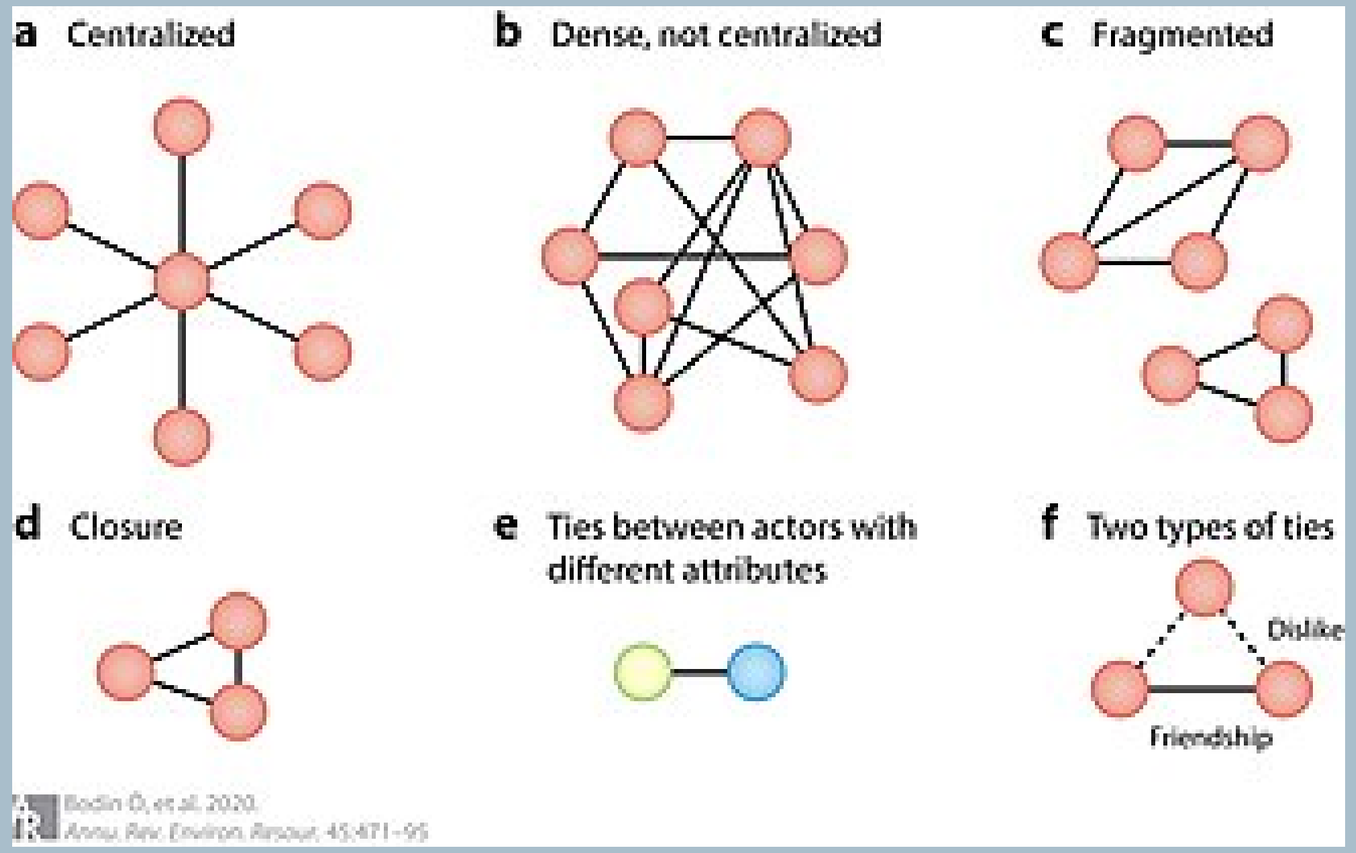
Social Network Theory Overview

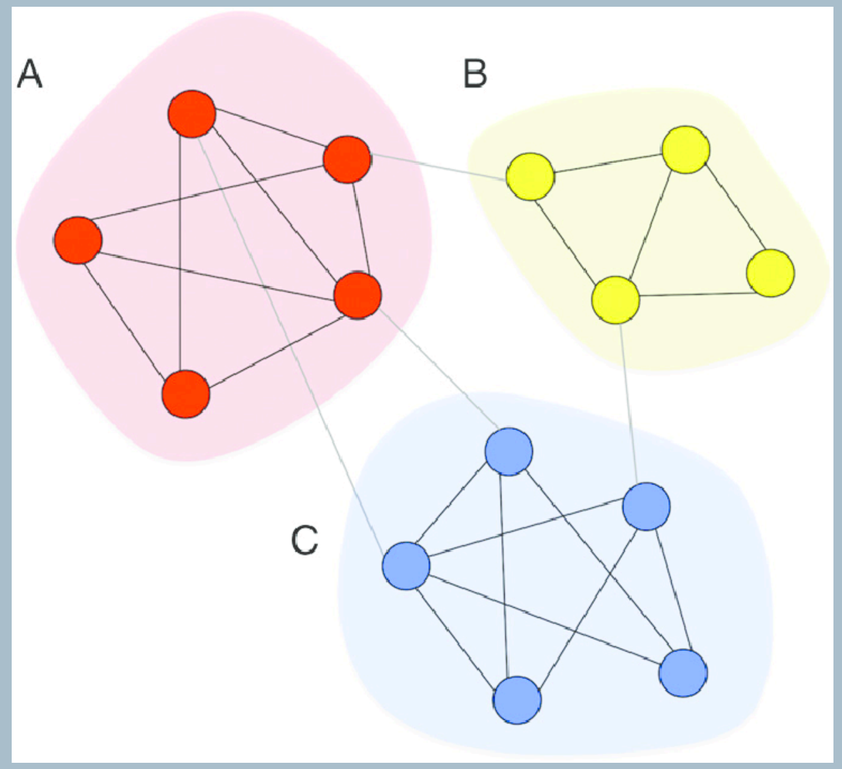
Social Network Theory examines how individuals are connected to one another through various types of relationships (e.g., friendship, kinship, professional). It analyzes the patterns and structures of these connections within social systems and their impact on individual behaviors and outcomes.

Key Concepts

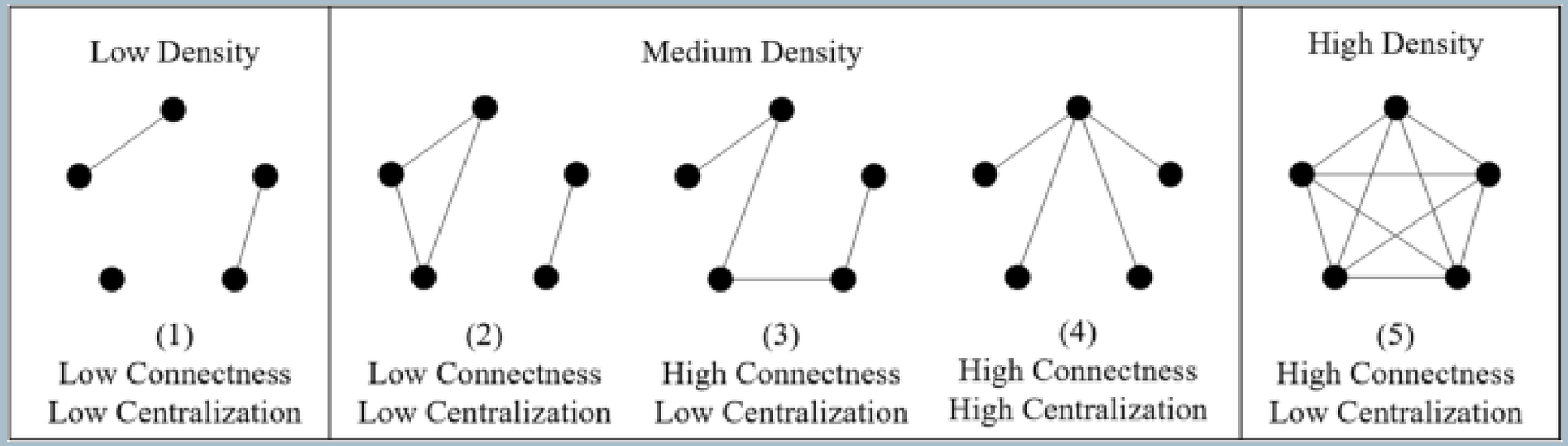
- **Nodes**: The individuals within the network.
- **Ties**: The connections between the individuals (can be strong or weak).
- **Centrality**: The importance or influence of a node/individual within the network.
- **Density**: The overall level of interconnectedness within the network.
- **Homophily**: The tendency of individuals to associate with similar others.

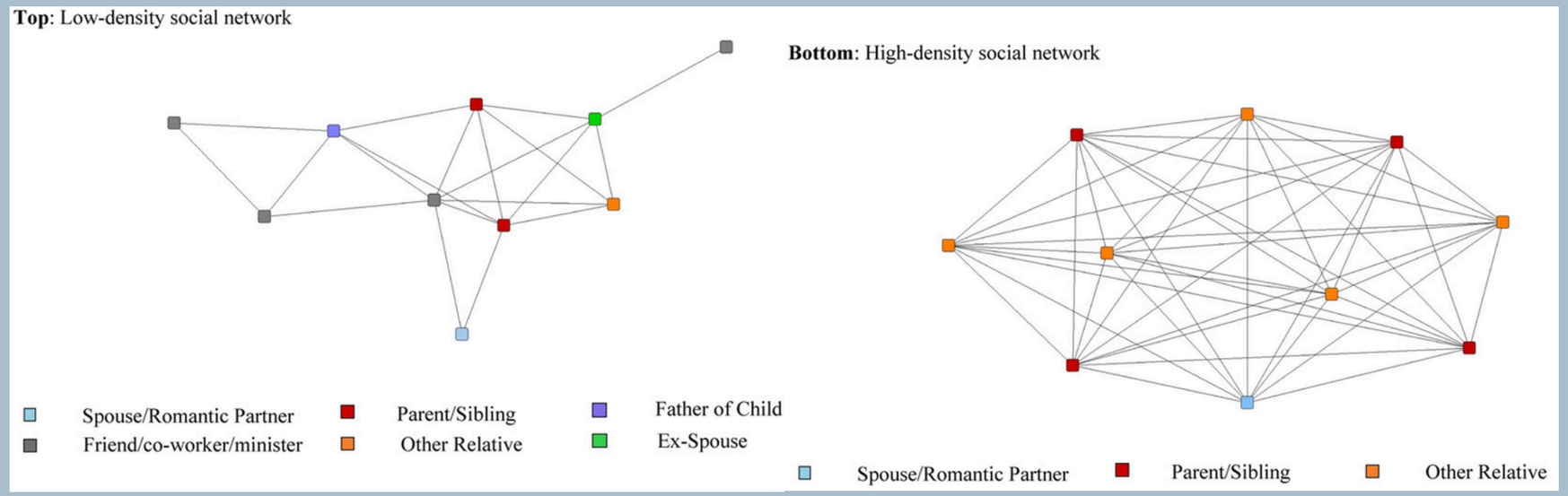
Centrality Nodes and Ties Homophily





Density


Low Density
Medium Density
High Density
(1)
Low Connectness
Low Centralization
(2)
Low Connectness
Low Centralization
(3)
High Connectness
Low Centralization
(4)
High Connectness
High Centralization
(5)
High Connectness
Low Centralization

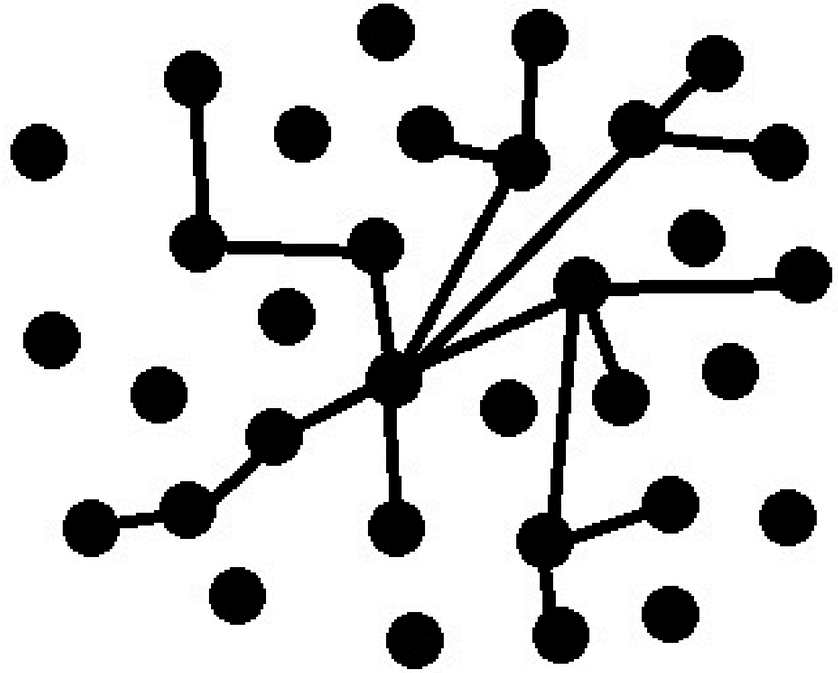

Top: Low-density social network
Bottom: High-density social network
Spouse/Romantic Partner
Parent/Sibling
Father of Child
Friend/co-worker/minister
Other Relative
Ex-Spouse
Spouse/Romantic Partner
Parent/Sibling
Other Relative

Characteristics of Isolation & Loneliness

Is Social Isolation & Loneliness Linked to Social Network Structure?

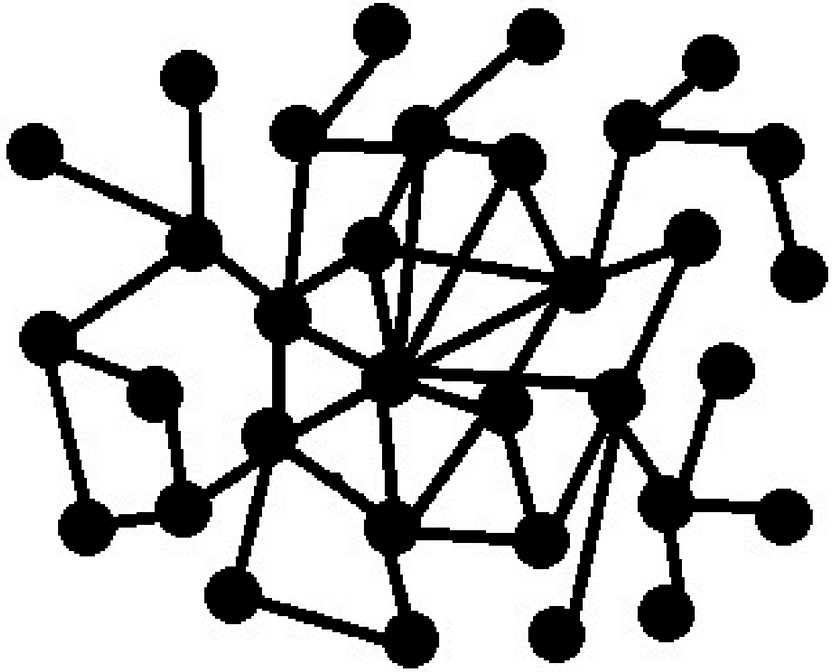
Key Influence: Network Density

- Network density refers to how well-connected the individuals within a network are to each other.
- **High Density:** In dense networks, most members (nodes) are interconnected, which helps strengthen social bonds and reduce feelings of loneliness.
- **Low Density:** Sparse networks, where members are less connected, can make individuals feel isolated, even in a crowd.
- Dense networks are associated with better social support and fewer feelings of isolation (Cacioppo, Fowler, & Christakis, 2009; Falci & McNeely, 2009). These types of networks can provide sense of community and belonging, which can alleviate feelings of loneliness.



weak network

unintegrated individuals, individuals held to network by only one relationship, network overreliant on star figures



strong network

all individuals integrated, many individuals held to network by multiple relationships, reliance on star figures diminished by relationships unrelated to star figures

Applying Social Network Theory

- Social network size (number of Nodes), social network cohesion (density) and loneliness may provide important insights
- helps reveal how well individuals are integrated on social media, which can influence feelings of loneliness.

- By examining the structure and dynamics of social networks, Social Network Theory provides a framework for understanding the mechanisms through which social media may contribute to or alleviate social isolation.

Significance

- **Addressing Identified Gaps:**
  - The research focuses on understanding how social media use affects social network structures and perceived social isolation.
  - Approach: Utilizes Social Network Theory and AI tools to provide a detailed analysis of these relationships.

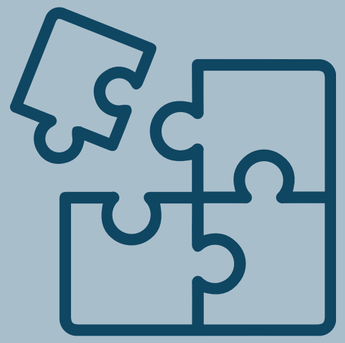
- **Broader Implications:**
  - Findings could inform public health initiatives aimed at reducing social isolation.
  - Insights could guide enhancements in social media platform designs to improve mental well-being among youth.

Objectives & Research Questions



Descriptive Research Questions

- How prevalent is perceived social isolation among adolescents and young adults who frequently use social media?
- What patterns of social media use are most strongly associated with feelings of loneliness and isolation?



Analytical Research Questions

- How does the frequency and type of social media use (e.g., passive vs. active) affect the structure of an individual's social network?
- What is the relationship between changes in social network density and centrality due to social media use and perceived social isolation?



Research Goals

- Examine how social media affects social network structure and quality among youth.
- Evaluate the effectiveness of strategies to enhance social connections and reduce isolation, including communication training and restrictions on passive social media use.

Main Research Question

RQ1 **How do patterns of social media use affect the structure and quality of social networks and perceived social isolation among youth?**

- Summary: Investigate the relationship between social media use, network structure (e.g., density, nodes/ties), and feelings of social isolation.
- Analysis Approach: Employ qualitative & quantitative measures to assess social connections and isolation levels.

Hypothesis



Hypothesis

- **Changes in social network structures due to social media use are associated with higher levels of perceived social isolation.**

Sub-hypotheses

- H1: A larger network size combined with low density (many connections but shallow interactions) on social media correlates with higher levels of perceived loneliness.
- H2: Higher frequency of passive social media use (such as scrolling through feeds without engaging) is associated with increased feelings of social isolation among adolescents.
- H3: Longer duration of social media use per day is linked to diminished quality of offline relationships, resulting in increased perceived loneliness.
- H4: Active social media use (posting and interacting) reduces loneliness through greater engagement. However, it also promotes homophily, as algorithms group users with similar views together, limiting diversity more than in offline settings.

Conceptual Framework

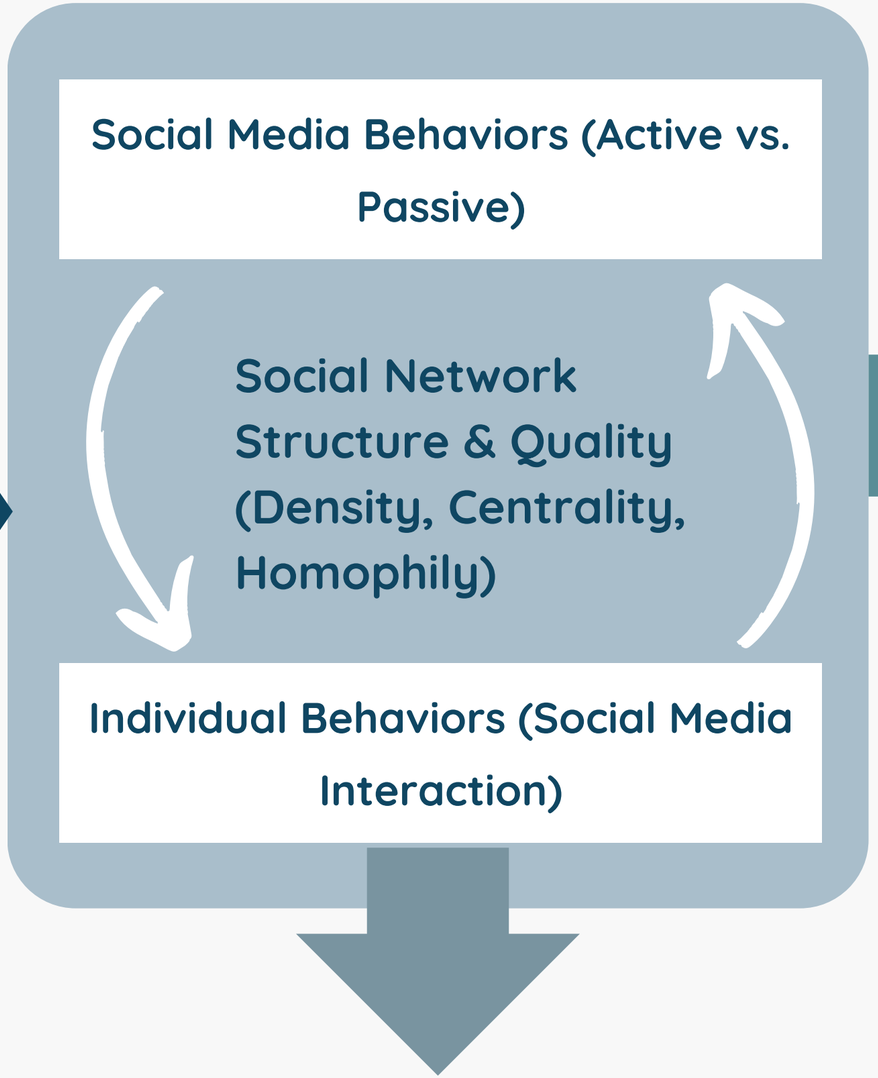
Things to Observe:

Frequency of Social Media Use

Types of Interactions

Quality and depth of interpersonal relationships

Perceived isolation



Reduced Social Isolation & Enhanced Social Networks

Strategies to Enhance Social Media Use for Better Social Connection and Meaningful Connections

Methodology: Study Overview

Cross-sectional & Mixed Methods Qualitative Approach:

- **Anonymous Survey Data Collection:** Gather data on social network structure, self-perceived isolation, and social media usage patterns.
- **Semi-Structured Interviews/Focus Groups:** To delve deeper into individual experiences and perceptions.
    -



Methodology: Qualitative

Data Collection Method

- Semi-Structured Interviews or Focus Groups
    - Conduct in-depth interviews or focus groups to explore the nuanced impact of social media on social connections and isolation.
- **Understanding Social Media Ecology:** Aim to delve into participants' perceptions and behaviors to better understand the role of social media in shaping social networks and sense of isolation.

Qualitative Questions: To understand Social Media Ecology

- Can you walk me through your day and describe how and when you interact with social media?
- Do you currently track the time you spend on social media? If not, would you be interested in learning how to track your screen time?
- How would you define overuse of social media?
- How do you use social media to socialize with friends? What specific activities do you engage in?
- What social media platforms do you use, and how do you use each of them?
- When you actively engage on social media (e.g., posting, commenting), how does it make you feel compared to when you are just scrolling or observing?
- How do you feel after using social media, particularly in terms of socializing with peers? Do you feel that these platforms strengthen or weaken those connections?
- Do you prefer interacting with close friends or acquaintances on social media? Why?
- What do you gain from using social media platforms?
- What do you feel you miss out on when you do not use social media? For example, how does it affect your social relationships? (e..g., Prob for gain/lose for social relationship)
- How do you feel about the time you spend on social media?

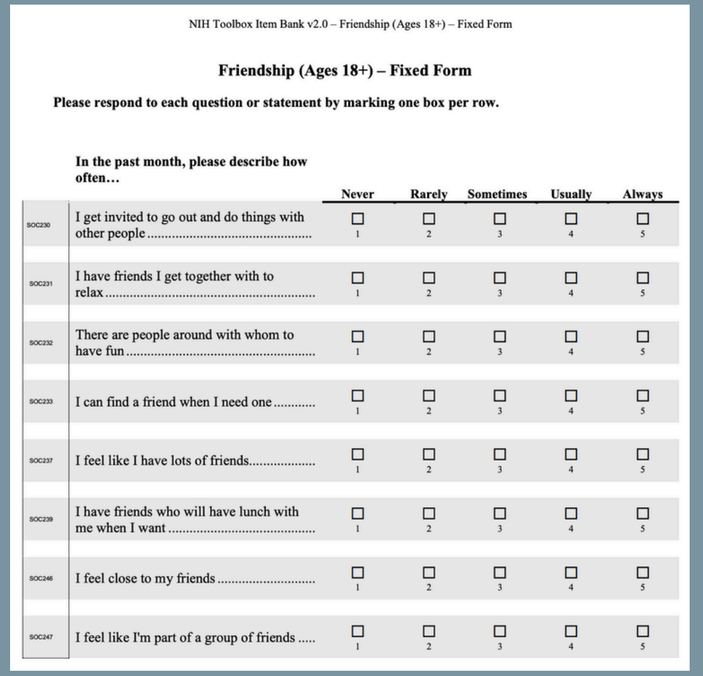
Methodology: Quantitative

- Conduct anonymous surveys to gather comprehensive data on social network structures, perceived isolation, and social media usage patterns.
- **Predictors:** Examine network size, density, social media use patterns.
- **Outcomes: S**elf-perceived loneliness as primary outcome.

Measurement Tools

- Network Analysis: Examine number of nodes/connections participant has within social network.
- Use validated scales to assess the quality/frequency of social interactions & measure feelings of social detachment.
- Gather data on Frequency/Duration of SMU, Motivations for Use, & types of social Activities (active vs. passive SMU)

Survey Instruments

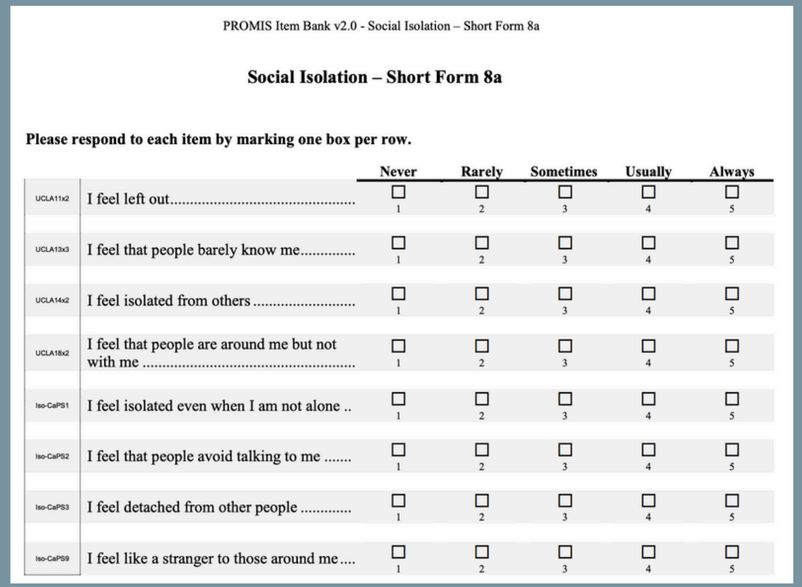


NIH Toolbox Item Bank v2.0 – Friendship (Ages 18+) – Fixed Form

Friendship (Ages 18+) – Fixed Form

Please respond to each question or statement by marking one box per row.

	In the past month, please describe how often…	Never	Rarely	Sometimes	Usually	Always
SOC230	I get invited to go out and do things with other people	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
SOC231	I have friends I get together with to relax	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
SOC232	There are people around with whom to have fun	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
SOC233	I can find a friend when I need one	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
SOC237	I feel like I have lots of friends	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
SOC239	I have friends who will have lunch with me when I want	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
SOC246	I feel close to my friends	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
SOC247	I feel like I'm part of a group of friends	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5



PROMIS Item Bank v2.0 - Social Isolation – Short Form 8a

Social Isolation – Short Form 8a

Please respond to each item by marking one box per row.

		Never	Rarely	Sometimes	Usually	Always
UCLA11x2	I feel left out	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
UCLA13x3	I feel that people barely know me	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
UCLA14x2	I feel isolated from others	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
UCLA18x2	I feel that people are around me but not with me	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
Iso-CaPS1	I feel isolated even when I am not alone	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
Iso-CaPS2	I feel that people avoid talking to me	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
Iso-CaPS3	I feel detached from other people	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
Iso-CaPS9	I feel like a stranger to those around me	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

Survey Instruments & Measurement Tools

- **TOOLBOX FF v2.0 - Emotional Support 8-17_11-17-2016**
  - Purpose: Assesses perceived emotional support, which is crucial for understanding the quality of social connections in adolescents.
- **Neuro-QoL Pediatric Short Form v1.0 - Social Relationships - Interaction with Peers 14Mar2014**
  - Purpose: Evaluates the quality and frequency of peer interactions, a vital aspect of social network engagement.
- **NIH Toolbox Pediatric Item Bank v3.0 - Friendship (Ages 8-17)_04-18-2023**
  - Purpose: Measures the strength and nature of friendships, providing insight into the presence of strong or weak social ties.
- **NIH Toolbox Pediatric Fixed Form v3.0 - Loneliness (Ages 8-17)_05-01-2023**
  - Purpose: Assesses feelings of loneliness, directly related to perceived social isolation.
- **PROMIS Item Bank v2.0 - Social Isolation 26Jan2015**
  - Purpose: Measures the extent of social isolation, a key variable in understanding the negative impacts of social network structures.
- **NIH Toolbox Friendship Fixed Form Age 18+ v2.0**
  - Purpose: Assess network density

Potential Challenges

Challenges:

Ethical Considerations: Ensuring confidentiality and anonymity, under 18 participation

Potential Solutions: Anonymized surveys, Adjusting for any confounding variables in the analysis

Thank You

Questions?